

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.38_k-1)

Metric

$$ds^2 = -dt \otimes dt + \left(-\frac{\left(\frac{\partial Y}{\partial r}\right)^2}{\epsilon f^2 + 1} \right) dr \otimes dr + Y^2 d\theta \otimes d\theta + Y^2 \sinh(\theta)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$f(r) = f$$

$$Y(r, t) = Y$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 + 1}} \frac{\partial}{\partial r} Y \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 + 1}} \frac{\partial}{\partial r} Y \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} Y \right) d\theta + \left(\frac{1}{2} i \sqrt{2} Y \sinh(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} Y \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} Y \sinh(\theta) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2} \sqrt{-\epsilon f^2 - 1}}{2 \frac{\partial}{\partial r} Y} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2} \epsilon f^2}{2 \sqrt{-\epsilon f^2 - 1} \frac{\partial}{\partial r} Y} + \frac{\sqrt{2}}{2 \sqrt{-\epsilon f^2 - 1} \frac{\partial}{\partial r} Y} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2 Y} \right) d\theta + \left(\frac{i \sqrt{2}}{2 Y \sinh(\theta)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 Y} \right) d\theta + \left(-\frac{i \sqrt{2}}{2 Y \sinh(\theta)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2} \sqrt{-\epsilon f^2 - 1}}{2 \frac{\partial}{\partial r} Y} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2} \epsilon f^2}{2 \sqrt{-\epsilon f^2 - 1} \frac{\partial}{\partial r} Y} + \frac{\sqrt{2}}{2 \sqrt{-\epsilon f^2 - 1} \frac{\partial}{\partial r} Y} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2 Y} \partial_\theta + \frac{i \sqrt{2}}{2 Y \sinh(\theta)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 Y} \partial_\theta - \frac{i \sqrt{2}}{2 Y \sinh(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\epsilon f(r)^2}{2\sqrt{-\epsilon f(r)^2 - 1}Y} + \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2Y} + \frac{\sqrt{2}}{2\sqrt{-\epsilon f(r)^2 - 1}Y}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}\epsilon f(r)^2}{2\sqrt{-\epsilon f(r)^2 - 1}Y} - \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2Y} + \frac{\sqrt{2}}{2\sqrt{-\epsilon f(r)^2 - 1}Y}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cosh(\theta)}{4Y\sinh(\theta)}$$

$$\beta = \frac{\sqrt{2}\cosh(\theta)}{4Y\sinh(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial^2}{\partial r\partial t}Y}{4\frac{\partial}{\partial r}Y}$$

$$\gamma = \frac{\sqrt{2} \frac{\partial^2}{\partial r \partial t} Y}{4 \frac{\partial}{\partial r} Y}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{\epsilon f(r) \frac{\partial}{\partial r} f(r)}{2 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{2 Y \frac{\partial}{\partial r} Y} - \frac{\frac{\partial^2}{(\partial t)^2} Y}{2 Y}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\epsilon f(r)^2}{4 Y^2} - \frac{\frac{\partial^3}{\partial r (\partial t)^2} Y}{4 \frac{\partial}{\partial r} Y} + \frac{\frac{\partial}{\partial t} Y^2}{4 Y^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{\epsilon f(r) \frac{\partial}{\partial r} f(r)}{2 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{2 Y \frac{\partial}{\partial r} Y} - \frac{\frac{\partial^2}{(\partial t)^2} Y}{2 Y}$$

$$\Lambda = \frac{\epsilon f(r)^2}{12 Y^2} + \frac{\epsilon f(r) \frac{\partial}{\partial r} f(r)}{6 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^3}{\partial r (\partial t)^2} Y}{12 \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{6 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial}{\partial t} Y^2}{12 Y^2} + \frac{\frac{\partial^2}{(\partial t)^2} Y}{6 Y}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\epsilon f\left(r\right)^2}{6Y^2} + \frac{\epsilon f\left(r\right)\frac{\partial}{\partial r}f\left(r\right)}{6Y\frac{\partial}{\partial r}Y} - \frac{\frac{\partial^3}{\partial r(\partial t)^2}Y}{6\frac{\partial}{\partial r}Y} + \frac{\frac{\partial^2}{\partial r\partial t}Y\frac{\partial}{\partial t}Y}{6Y\frac{\partial}{\partial r}Y} - \frac{\frac{\partial}{\partial t}Y^2}{6Y^2} + \frac{\frac{\partial^2}{(\partial t)^2}Y}{6Y}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"